



Topic: "The Ego Empire" – CEO Narcissism and Corporate Expansion

FINANCIAL ECONOMETRICS

Members:

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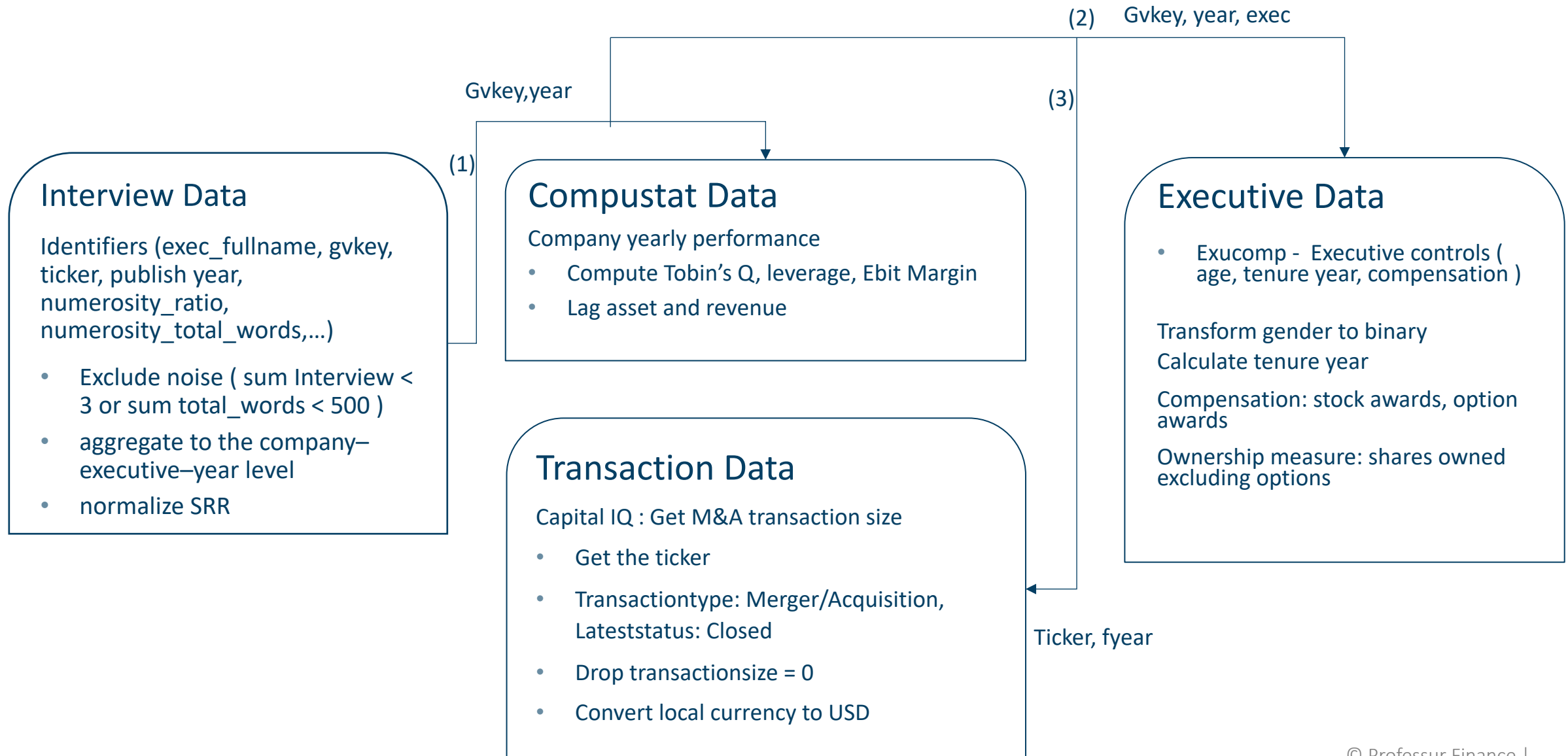


- **Research Question:** Does CEO narcissism drive excessive “Empire Building” ? Are narcissistic executives more likely to pursue aggressive M&A activities and asset growth, prioritizing size and status over operational efficiency?



- **Hypothesis:** Executives with higher narcissism scores will be positively associated with higher M&A spending

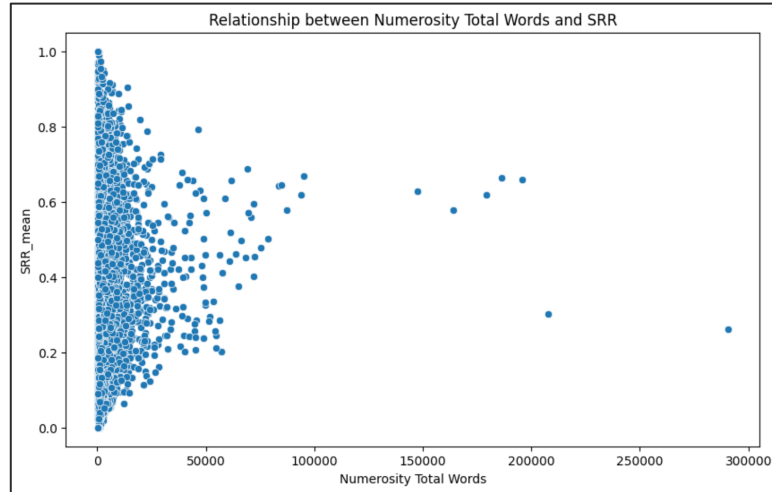
Data Preparation



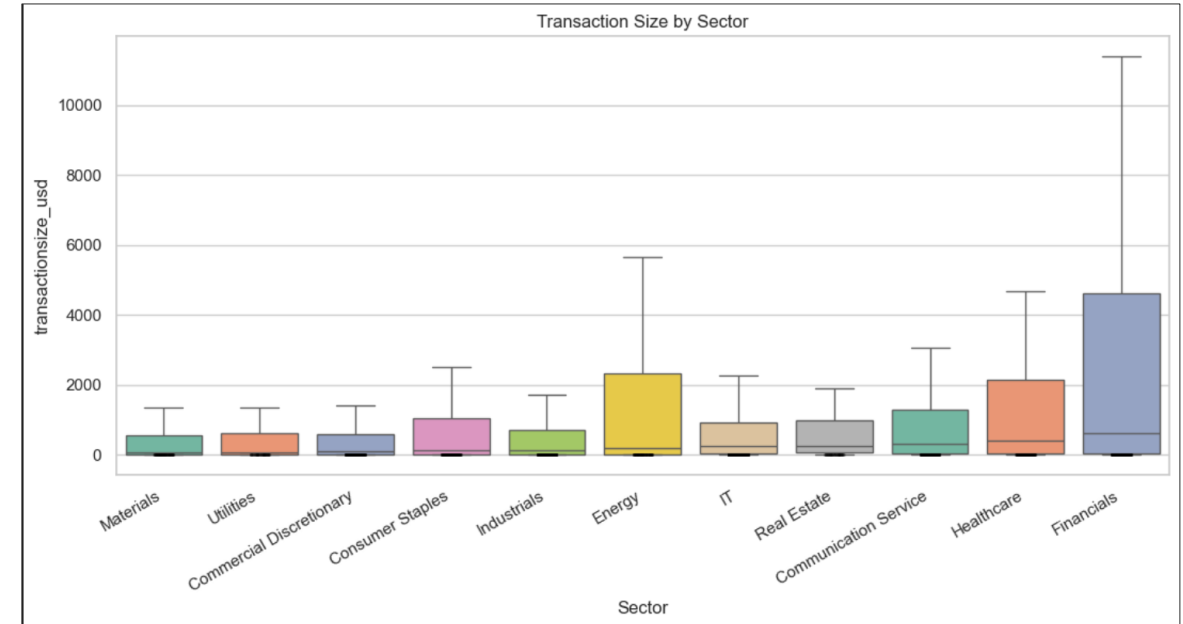
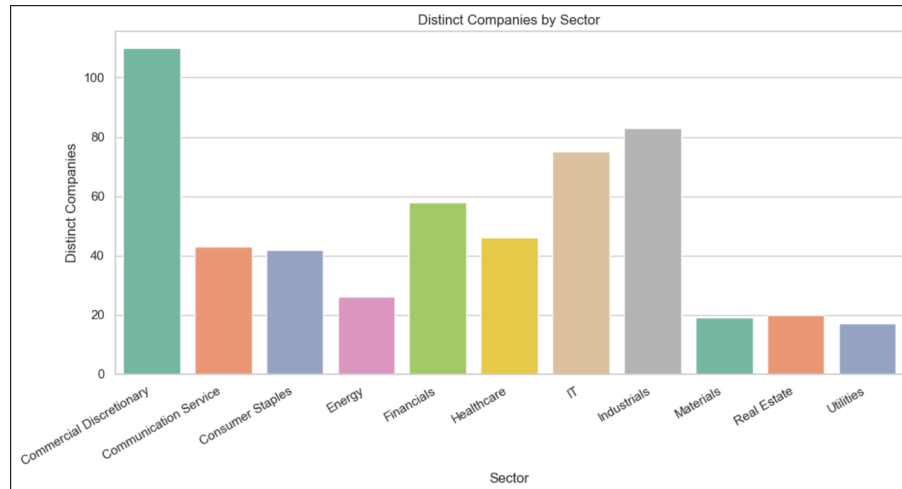


	Number of Observations	Unique firms	Unique CEOs	Year
Initial Interview Data set	24255	1106	2259	2005-2025
Interview Data Set after transformation	6508	921	1835	2007-2025
Compustat after transformation	15711	954		2005-2025
Executive after transformation	89417	966	16675	2005-2025
Transaction Data after transformation	12926	907		2005-2025
Merged Data(exclude nan value of transactionsize, exec_fullname, and TobinQ)	2567	539	849	2007-2024
Final Data (exclude nan values of all variables used in model)	965	222	266	2007-2024

Exploratory Data



short interviews show high SRR -> SRR should be normalized



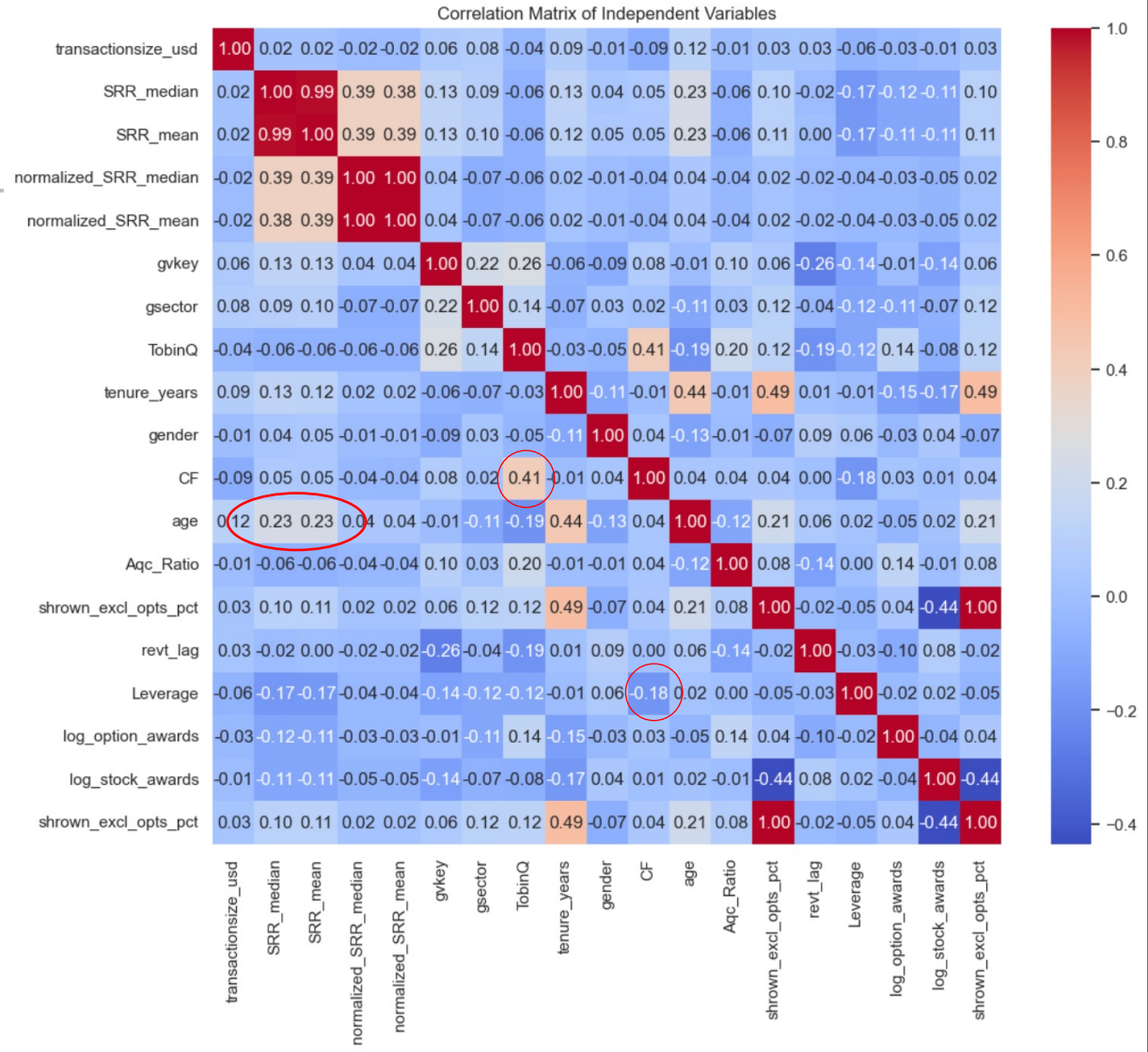
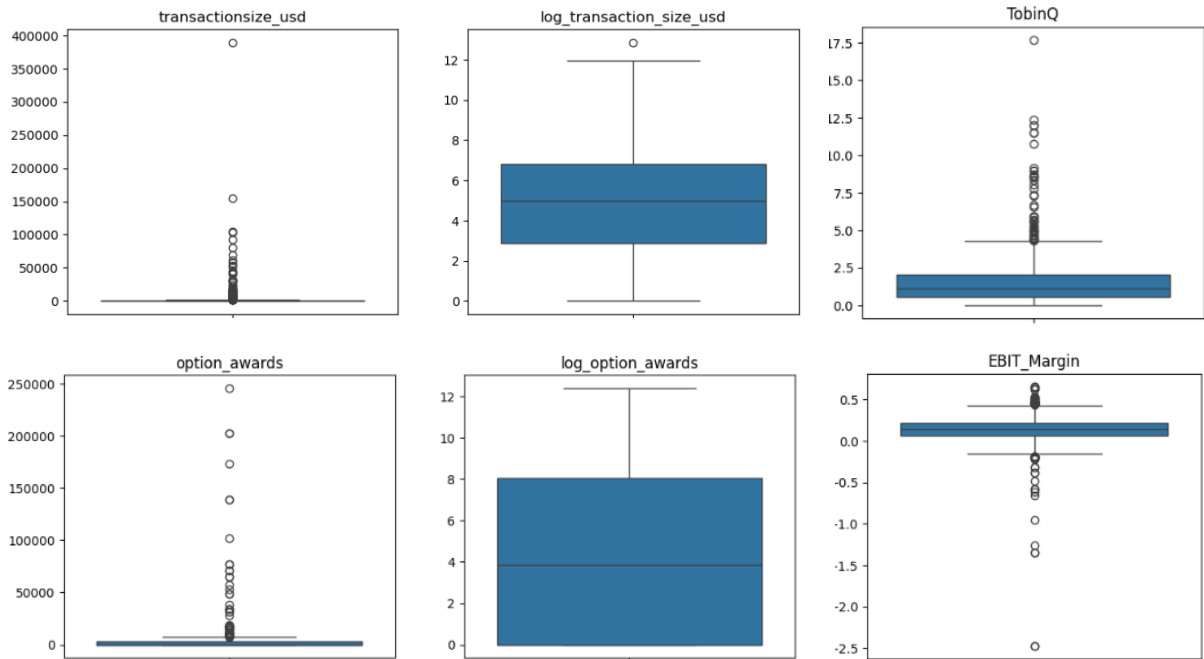
Financials, Healthcare and Energy have the largest typical deal sizes.

Commercial Discretionary dominate, followed by IT, Financials and Industrials

- aggregate deal activity is driven mostly by these high-volume industries
- Motivate industry controls or fixed effect: sector

Exploratory Data

	transactionsize_usd	revt_lag	CF	bonus	option_awards	stock_awards
count	2567.000000	2564.000000	2567.000000	2567.000000	2567.000000	2567.000000
mean	2751.656857	34993.829395	0.101530	431.997590	2574.869905	7741.525548
std	14670.505491	63422.949882	0.076860	1511.916572	11649.249853	15352.946273
min	0.000000	10.029000	-0.331062	0.000000	0.000000	0.000000
25%	22.862110	3889.000000	0.056567	0.000000	0.000000	2397.715000
50%	188.136399	10701.000000	0.093801	0.000000	0.000000	5280.000000
75%	1125.000000	36216.000000	0.137867	0.000000	2195.400000	9559.842000
max	389694.211213	645737.000000	0.574269	19350.000000	245953.429000	394592.295000



Most correlations are small (close to 0)

right-skewed data (outliners):
transactionsize_usd, revt, at,
option_awards, stock_awards



Result : a firm-year dataset combining (i) narcissism proxies (normalized SRR), (ii) M&A “empire building” outcomes, (iii) firm financial controls, and (iv) CEO characteristics for regression analysis.

- **TobinQ** : growth opportunity, high growth opportunities firms are more potential to expand their size and do more deals.
- **EBIT_Margin**: Profitable firms (high EBITDA margin) have stronger internal funds for deals
- **Leverage**: high debt can limit new acquisitions.
- **log_revt_lag**: Sales power
- **log_at_lag**: acquiror' total asset last fiscal year, Larger firms have more resources for M&A

- **tenure_years, gender, age**: influence risk tolerance and deal initiation; older/tenured CEOs may be more conservative
- **shrown_excl_opts_pct**: stock ownership
- **log_stock_awards**: compensation package of CEO
- **log_option_awards** : compensation package of CEO

- ☐ Simple model, model with controls, models with controls and fixed effect
- ☐ Industry * SRR interaction model
- ☐ Fixed effect : firm, sector, year

Same year Regression – simple model and control model



Simple regression on log_transaction_size_usd, nonrobust:

	coef	std_err	p_value	t_stat	R_squared	adj_R_squared	n
model							
SRR_median	0.669448	0.420968	0.112104	1.590260	0.002619	0.001584	965
SRR_mean	0.816221	0.432003	0.059140	1.889386	0.003693	0.002659	965
normalized_SRR_mean	-66.782098	74.455089	0.369973	-0.896945	0.000835	-0.000203	965
normalized_SRR_median	-67.594356	74.425515	0.363992	-0.908215	0.000856	-0.000182	965

Regression with Controls Results on log_transaction_size_usd with Controls:, no clustering

	Independent_Variable	Coefficient	Std_Error	T_Stat	P_Value	[0.025	0.975]	R_Squared
0	SRR_median	0.270576	0.414891	0.652161	0.514455	-0.543632	1.084783	0.110410
1	SRR_mean	0.356056	0.426407	0.835015	0.403919	-0.480750	1.192862	0.110664
2	normalized_SRR_mean	-19.862534	71.243908	-0.278796	0.780462	-159.675781	119.950712	0.110085
3	normalized_SRR_median	-20.260794	71.218074	-0.284489	0.776097	-160.023343	119.501755	0.110088

Regression with Controls Results on log_transaction_size_usd (clustered by gvkey):

	Independent_Variable	coef	Std_Error	T_Stat	P_Value	[0.025	0.975]	R_Squared	Adj_R_Squared	N_Clusters
0	SRR_median	0.270576	0.393389	0.687807	0.491574	-0.500453	1.041605	0.110410	0.099196	222
1	SRR_mean	0.356056	0.400473	0.889089	0.373955	-0.428856	1.140968	0.110664	0.099453	222
2	normalized_SRR_mean	-19.862534	32.193188	-0.616979	0.537248	-82.960024	43.234956	0.110085	0.098867	222
3	normalized_SRR_median	-20.260794	32.277204	-0.627712	0.530193	-83.522952	43.001364	0.110088	0.098870	222

$$\ln(\text{TransactionSize})_{it} = \beta_0 + \beta_1 \text{SRR}_{it} + \sum_{k=1}^n \gamma_k \text{Controls}_{it,k} + \epsilon_{it}$$

Non-robust model and robust model with clustered SE by firm

Components:

y = log(transaction_size)

X = SRR (SRR_mean/median , normalized SRR mean/median)

Controls: TobinQ , tenure years, gender, age, stock awards , EBIT margin, option awards, leverage, Total Shares Owned, revenue, asset (some variables in log transformation)

Result:

- SRR does not significantly explain current-year acquisition transaction
- Non-robust model is consistent with robust model

Same year Regression – simple model and control model



OLS Regression Results						
=====						
Dep. Variable:	log_transaction_size_usd	R-squared:	0.110			
Model:	OLS	Adj. R-squared:	0.099			
Method:	Least Squares	F-statistic:	9.814			
Date:	Fri, 30 Jan 2026	Prob (F-statistic):	2.87e-18			
Time:	23:13:53	Log-Likelihood:	-2266.5			
No. Observations:	965	AIC:	4559.			
Df Residuals:	952	BIC:	4622.			
Df Model:	12					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	0.0286	0.907	0.031	0.975	-1.752	1.809
normalized_SRR_median	-20.2608	71.218	-0.284	0.776	-160.023	119.502
TobinQ	0.1596	0.055	2.921	0.004	0.052	0.267
tenure_years	0.0029	0.013	0.224	0.823	-0.023	0.028
gender	0.4373	0.343	1.273	0.203	-0.237	1.111
age	0.0127	0.015	0.839	0.402	-0.017	0.042
log_stock_awards	-0.0020	0.033	-0.061	0.951	-0.067	0.063
EBIT_Margin	1.2352	0.428	2.888	0.004	0.396	2.075
shrown_excl_opts_pct	0.0213	0.025	0.861	0.390	-0.027	0.070
Leverage	-0.3417	0.461	-0.741	0.459	-1.246	0.563
log_option_awards	-0.0438	0.021	-2.096	0.036	-0.085	-0.003
log_revt_lag	-0.1914	0.112	-1.712	0.087	-0.411	0.028
log_at_lag	0.5742	0.100	5.728	0.000	0.377	0.771

- Normalized SRR is not significant

Significant controls and positive coefficient:

- **Tobin's Q:** High-growth firms spend more on deals.
- **EBIT Margin:** Higher profitability leads to larger deals
-> Firms don't just need to be big; they need to be efficient to justify larger deals.
- **Assets:** Firm size is a primary driver of deal scale. This is the most reliable predictor. Since the variable is logged, this suggests a power law relationship: as a firm's total assets increase, their deal capacity grows substantially -> Larger firms have the balance sheets to support bigger tickets.

Same year Regression – with FE



$$\ln(\text{TransactionSize})_{it} = \beta_0 + \beta_1 \text{SRR}_{it} + \sum_{k=1}^n \gamma_k \text{Controls}_{it,k} + \alpha_{it} + \delta_t + \epsilon_{it}$$

Regression with Controls and Fixed Effects Results on log_transaction_size_usd:,

	FE_spec	Independent_Variable	Coefficient	Std_Error	T_Stat	P_Value
0	FE1: sector	SRR_median	-0.045327	0.423328	-0.107074	0.914753
1	FE1: sector	SRR_mean	0.000926	0.436059	0.002123	0.998306
2	FE1: sector	normalized_SRR_mean	-17.675723	71.336462	-0.247780	0.804359
3	FE1: sector	normalized_SRR_median	-17.853952	71.303894	-0.250392	0.802339
4	FE2: sector + year	SRR_median	0.021914	0.437456	0.050095	0.960058
5	FE2: sector + year	SRR_mean	0.116297	0.451821	0.257396	0.796930
6	FE2: sector + year	normalized_SRR_mean	3.167614	72.077478	0.043947	0.964956
7	FE2: sector + year	normalized_SRR_median	2.578818	72.038100	0.035798	0.971451
8	FE3: firm	SRR_median	-0.447099	0.561614	-0.796096	0.426235
9	FE3: firm	SRR_mean	-0.294058	0.578925	-0.507937	0.611651
10	FE3: firm	normalized_SRR_mean	-46.241982	83.918686	-0.551033	0.581779
11	FE3: firm	normalized_SRR_median	-46.466175	83.909304	-0.553767	0.579908
12	FE4: sector + firm	SRR_median	-0.447099	0.561614	-0.796096	0.426235
13	FE4: sector + firm	SRR_mean	-0.294058	0.578925	-0.507937	0.611651
14	FE4: sector + firm	normalized_SRR_mean	-46.241982	83.918686	-0.551033	0.581779
15	FE4: sector + firm	normalized_SRR_median	-46.466175	83.909304	-0.553767	0.579908

➤ No evidence to conclude SRR is associated with current-year acquisition transaction

Regression with Controls and Fixed Effects (clustered by firm gvkey):

	FE_spec	Independent_Variable	Coefficient	Std_Error	T_Stat	P_Value
0	FE1: sector	SRR_median	-0.045327	0.430679	-0.105246	0.916181
1	FE1: sector	SRR_mean	0.000926	0.443336	0.002089	0.998334
2	FE1: sector	normalized_SRR_mean	-17.675723	32.813193	-0.538677	0.590110
3	FE1: sector	normalized_SRR_median	-17.853952	32.862525	-0.543292	0.586929
4	FE2: sector + year	SRR_median	0.021914	0.425172	0.051542	0.958894
5	FE2: sector + year	SRR_mean	0.116297	0.435664	0.266942	0.789514
6	FE2: sector + year	normalized_SRR_mean	3.167614	30.464737	0.103976	0.917188
7	FE2: sector + year	normalized_SRR_median	2.578818	30.560194	0.084385	0.932750
8	FE3: firm	SRR_median	-0.447099	0.488698	-0.914877	0.360256
9	FE3: firm	SRR_mean	-0.294058	0.515950	-0.569934	0.568722
10	FE3: firm	normalized_SRR_mean	-46.241982	41.917482	-1.103167	0.269955
11	FE3: firm	normalized_SRR_median	-46.466175	41.909133	-1.108736	0.267544
12	FE4: sector + firm	SRR_median	-0.447099	0.492075	-0.908598	0.363562
13	FE4: sector + firm	SRR_mean	-0.294058	0.519516	-0.566023	0.571378
14	FE4: sector + firm	normalized_SRR_mean	-46.241982	42.207171	-1.095595	0.273256
15	FE4: sector + firm	normalized_SRR_median	-46.466175	42.198764	-1.101126	0.270842

Next year Regression – simple model and model with controls



Simple regression on log_transaction_size_usd_nextyear, nonrobust:

	coef	std_err	p_value	t_stat	R_squared
model					
SRR_median	1.236708	0.419748	0.003293	2.946311	0.008934
SRR_mean	1.327601	0.430803	0.002117	3.081691	0.009765
normalized_SRR_mean	119.290238	74.407334	0.109217	1.603205	0.002662
normalized_SRR_median	119.768814	74.377687	0.107665	1.610279	0.002685

Simple regression on log_transaction_size_usd_nextyear, firm-clustered

	coef	std_err	p_value	t_stat	R_squared
model					
SRR_median	1.236708	0.497381	0.012903	2.486443	0.008934
SRR_mean	1.327601	0.508296	0.009005	2.611864	0.009765
normalized_SRR_mean	119.290238	59.052315	0.043375	2.020077	0.002662
normalized_SRR_median	119.768814	59.169189	0.042952	2.024175	0.002685

Regression with Controls Results on log_transaction_size_usd_ne

	Independent_Variable	Coefficient	Std_Error	T_Stat	P_Value
0	SRR_median	0.937836	0.419153	2.237457	0.025487
1	SRR_mean	0.964728	0.430846	2.239147	0.025377
2	normalized_SRR_mean	165.887062	71.950863	2.305560	0.021349
3	normalized_SRR_median	166.735987	71.922687	2.318267	0.020646

No clustering

Regression with Controls Results on log_transaction_size_usd_ne

	Independent_Variable	coef	Std_Error	T_Stat	P_Value
0	SRR_median	0.937836	0.420967	2.227812	0.025893
1	SRR_mean	0.964728	0.429879	2.244185	0.024821
2	normalized_SRR_mean	165.887062	69.157706	2.398678	0.016454
3	normalized_SRR_median	166.735987	69.295991	2.406142	0.016122

Firm-clustered

All SRR in 2 models are statistically significant except for normalized SRR in non-robust model

All SRR are statistically significant in model with controls

-> One possible hypothesis is that SRR has a lagged effect. It takes time to turn into a boardroom decision for a merger.

Next year Regression clustered by Firm



OLS Regression Results

Dep. Variable:	log_transaction_size_usd_nextyear	R-squared:	0.089
Model:	OLS	Adj. R-squared:	0.078
Method:	Least Squares	F-statistic:	8.442
Date:	Fri, 30 Jan 2026	Prob (F-statistic):	2.58e-14
Time:	23:13:54	Log-Likelihood:	-2278.2
No. Observations:	965	AIC:	4580.
Df Residuals:	953	BIC:	4639.
Df Model:	11		

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.6213	0.918	0.677	0.499	-1.180	2.422
normalized_SRR_mean	168.8879	72.058	2.344	0.019	27.477	310.299
TobinQ	0.1332	0.055	2.417	0.016	0.025	0.241
tenure_years	0.0066	0.013	0.513	0.608	-0.019	0.032
gender	0.4190	0.347	1.209	0.227	-0.261	1.099
age	-0.0027	0.015	-0.180	0.857	-0.033	0.027
log_stock_awards	-0.0092	0.034	-0.275	0.784	-0.075	0.057
EBIT_Margin	0.9462	0.432	2.189	0.029	0.098	1.795
shrown_excl_opts_pct	0.0245	0.025	0.979	0.328	-0.025	0.073
Leverage	-0.2047	0.466	-0.439	0.661	-1.120	0.710
log_rev_lag	-0.1093	0.111	-0.985	0.325	-0.327	0.108
log_at_lag	0.5114	0.100	5.128	0.000	0.316	0.707

OLS Regression Results

Dep. Variable:	log_transaction_size_usd_nextyear	R-squared:	0.093
Model:	OLS	Adj. R-squared:	0.081
Method:	Least Squares	F-statistic:	5.295
Date:	Fri, 30 Jan 2026	Prob (F-statistic):	7.65e-08
Time:	23:13:54	Log-Likelihood:	-2276.0
No. Observations:	965	AIC:	4578.
Df Residuals:	952	BIC:	4641.
Df Model:	12		
Covariance Type:	cluster		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.6464	1.017	0.636	0.525	-1.347	2.640
normalized_SRR_median	166.7360	69.296	2.406	0.016	30.918	302.554
TobinQ	0.1424	0.059	2.411	0.016	0.027	0.258
tenure_years	0.0009	0.020	0.046	0.963	-0.039	0.040
gender	0.3701	0.295	1.256	0.209	-0.207	0.948
age	-0.0003	0.016	-0.017	0.986	-0.032	0.031
log_stock_awards	-0.0122	0.040	-0.302	0.763	-0.091	0.067
EBIT_Margin	0.9859	0.409	2.410	0.016	0.184	1.788
shrown_excl_opts_pct	0.0292	0.030	0.974	0.330	-0.030	0.088
Leverage	-0.2165	0.586	-0.369	0.712	-1.366	0.933
log_option_awards	-0.0434	0.023	-1.876	0.061	-0.089	0.002
log_rev_lag	-0.0633	0.163	-0.388	0.698	-0.383	0.256
log_at_lag	0.4736	0.148	3.200	0.001	0.184	0.764

Normalized SRR is significant.

Other controls stay consistent with model in same years.

(TobinQ, Ebit margin, asset positively coefficient with acquisition and statistically significant)

Next year Regression - with Fixed Effect



$$\ln(TransactionSize)_{it} = \beta_0 + \beta_1 SRR_{it} + \sum_{k=1}^n \gamma_k Controls_{it,k} + \alpha_{it} + \delta_t + \epsilon_{it}$$

- **normalized_SRR** is significant in all models

Key findings:

- Individual narcissism is associated with larger deals
- CEOs set M&A scale based on their communication style well before deals are announced (SRR has a lagged time effect. SRR in same year model is not significant but SRR in next year is significant)

	FE_spec	Independent_Variable	Coefficient	Std_Error	T_Stat	P_Value	[0.025	0.975]	R_Squared
0	FE1: sector	SRR_median	0.657448	0.427284	1.538669	0.124221	-0.181090	1.495987	0.112743
1	FE1: sector	SRR_mean	0.640093	0.440191	1.454126	0.146245	-0.223775	1.503961	0.112505
2	FE1: sector	normalized_SRR_mean	172.317030	71.876537	2.397403	0.016706	31.260368	313.373693	0.115908
3	FE1: sector	normalized_SRR_median	173.368918	71.840886	2.413235	0.016002	32.382219	314.355616	0.115979
4	FE2: sector + year	SRR_median	0.709219	0.443496	1.599157	0.110127	-0.161154	1.579593	0.123181
5	FE2: sector + year	SRR_mean	0.705351	0.458121	1.539662	0.123984	-0.193724	1.604427	0.123005
6	FE2: sector + year	normalized_SRR_mean	175.404662	72.946104	2.404579	0.016387	32.245807	318.563516	0.126216
7	FE2: sector + year	normalized_SRR_median	176.367161	72.903469	2.419187	0.015747	33.291979	319.442343	0.126282
8	FE3: firm	SRR_median	0.269903	0.569238	0.474147	0.635536	-0.847634	1.387439	0.409927
9	FE3: firm	SRR_mean	0.329885	0.586597	0.562370	0.574036	-0.821731	1.481500	0.410001
10	FE3: firm	normalized_SRR_mean	171.305315	84.815470	2.019741	0.043775	4.794354	337.816276	0.413022
11	FE3: firm	normalized_SRR_median	171.410135	84.805821	2.021207	0.043622	4.918115	337.902154	0.413026
12	FE4: sector + firm	SRR_median	0.269903	0.569238	0.474147	0.635536	-0.847634	1.387439	0.409927
13	FE4: sector + firm	SRR_mean	0.329885	0.586597	0.562370	0.574036	-0.821731	1.481500	0.410001
14	FE4: sector + firm	normalized_SRR_mean	171.305315	84.815470	2.019741	0.043775	4.794354	337.816276	0.413022
15	FE4: sector + firm	normalized_SRR_median	171.410135	84.805821	2.021207	0.043622	4.918115	337.902154	0.413026
16	FE5: year	SRR_median	0.992681	0.437069	2.271223	0.023360	0.134933	1.850429	0.102804
17	FE5: year	SRR_mean	1.036224	0.450456	2.300387	0.021645	0.152203	1.920245	0.102931
18	FE5: year	normalized_SRR_mean	165.682358	73.034411	2.268552	0.023523	22.352204	309.012512	0.102792

Next year Regression with FE: Sector



OLS Regression Results

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=====
Dep. Variable:    log_transaction_size_usd_nextyear    R-squared:    0.116
Model:            OLS                                Adj. R-squared: 0.095
Method:           Least Squares                      F-statistic:   4.153
Date:             Fri, 30 Jan 2026                    Prob (F-statistic): 1.75e-08
Time:             23:13:55                            Log-Likelihood: -2263.6
No. Observations: 965                                AIC:          4573.
Df Residuals:     942                                BIC:          4685.
Df Model:         22
Covariance Type:  cluster
=====

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	0.2370	1.143	0.207	0.836	-2.004	2.478
C(gsector)[T.15.0]	-0.0178	0.464	-0.038	0.969	-0.927	0.891
C(gsector)[T.20.0]	0.1209	0.480	0.252	0.801	-0.819	1.061
C(gsector)[T.25.0]	0.1154	0.469	0.246	0.806	-0.804	1.035
C(gsector)[T.30.0]	-0.2391	0.457	-0.523	0.601	-1.135	0.657
C(gsector)[T.35.0]	0.7938	0.522	1.520	0.128	-0.229	1.817
C(gsector)[T.40.0]	1.2360	0.590	2.093	0.036	0.079	2.393
C(gsector)[T.45.0]	0.6587	0.456	1.444	0.149	-0.236	1.553
C(gsector)[T.50.0]	1.2999	0.411	3.163	0.002	0.494	2.105
C(gsector)[T.55.0]	-0.4347	0.705	-0.616	0.538	-1.817	0.948
C(gsector)[T.60.0]	0.7114	0.815	0.873	0.383	-0.886	2.309
normalized_SRR_median	173.3689	62.141	2.790	0.005	51.575	295.163
TobinQ	0.1279	0.056	2.298	0.022	0.019	0.237
tenure_years	0.0037	0.019	0.197	0.844	-0.033	0.040
gender	0.3188	0.294	1.085	0.278	-0.257	0.895
age	0.0023	0.016	0.140	0.889	-0.030	0.034
log_stock_awards	-0.0049	0.040	-0.122	0.903	-0.084	0.074
EBIT_Margin	0.7350	0.416	1.768	0.077	-0.080	1.550
shrown_excl_opts_pct	0.0148	0.029	0.504	0.614	-0.043	0.072
Leverage	0.3667	0.600	0.611	0.541	-0.809	1.542
log_option_awards	-0.0385	0.023	-1.670	0.095	-0.084	0.007
log_revt_lag	0.1659	0.203	0.819	0.413	-0.231	0.563
log_at_lag	0.2205	0.183	1.205	0.228	-0.138	0.579

Normalized SRR is significant

Only Financials (gsector = 40) and Communication Service (gsector = 50) have significant positive coefficient

- Financial firms are more likely to engage in M&A activities and their SRR scores may reflect their strategic communication related to such activities.

Next year regression – SRR*sector as narcissism-industry interaction



	coef	std err	z	P> z
Intercept	0.4992	0.925	0.539	0.590
C(gsector_name)[T.Materials]	-0.1965	0.431	-0.456	0.649
C(gsector_name)[T.Industrials]	-0.2226	0.465	-0.479	0.632
C(gsector_name)[T.Commercial Discretionary]	-0.0924	0.437	-0.211	0.833
C(gsector_name)[T.Consumer Staples]	-0.5587	0.476	-1.173	0.241
C(gsector_name)[T.Healthcare]	0.5495	0.541	1.017	0.309
C(gsector_name)[T.Financials]	1.0100	0.646	1.563	0.118
C(gsector_name)[T.IT]	0.3133	0.444	0.705	0.481
C(gsector_name)[T.Communication Service]	0.9030	0.415	2.174	0.030
C(gsector_name)[T.Utilities]	-0.3120	0.809	-0.386	0.700
C(gsector_name)[T.Real Estate]	1.4157	0.795	1.781	0.075
normalized_SRR_median	-113.8639	36.435	-3.125	0.002
normalized_SRR_median:C(gsector_name)[T.Materials]	252.3609	301.812	0.836	0.403
normalized_SRR_median:C(gsector_name)[T.Industrials]	597.9551	163.406	3.659	0.000
normalized_SRR_median:C(gsector_name)[T.Commercial Discretionary]	266.8777	41.864	6.375	0.000
normalized_SRR_median:C(gsector_name)[T.Consumer Staples]	683.7818	626.805	1.091	0.275
normalized_SRR_median:C(gsector_name)[T.Healthcare]	358.6223	148.552	2.414	0.016
normalized_SRR_median:C(gsector_name)[T.Financials]	371.3690	413.149	0.899	0.369
normalized_SRR_median:C(gsector_name)[T.IT]	799.5035	206.039	3.880	0.000
normalized_SRR_median:C(gsector_name)[T.Communication Service]	1597.5113	768.723	2.078	0.038
normalized_SRR_median:C(gsector_name)[T.Utilities]	-1260.9793	2425.867	-0.520	0.603
normalized_SRR_median:C(gsector_name)[T.Real Estate]	-1592.8423	423.305	-3.763	0.000

	coef	std err	z	P> z
TobinQ	0.1311	0.056	2.356	0.018
tenure_years	0.0035	0.018	0.200	0.841
gender	0.2779	0.291	0.954	0.340
log_stock_awards	-0.0069	0.040	-0.174	0.862
EBIT_Margin	0.6509	0.404	1.610	0.107
shrown_excl_opts_pct	0.0134	0.029	0.464	0.643
Leverage	0.4185	0.605	0.692	0.489
log_option_awards	-0.0340	0.022	-1.530	0.126
log_rev_t_lag	0.1585	0.197	0.805	0.421
log_at_lag	0.2343	0.182	1.290	0.197

The SRR interact with sector coefficient is significantly in **Industrials, Consumer Discretionary, IT, Communication Service, Real Estate, Healthcare (based sector:Energy)**

normalized SRR is statistically significant (p_value < 0.001), p_value of SRR in this sub_industry model is more significant compared to previous model as it's p_value is very small

Next 2 year regression



Regression with Controls Results on log_transaction_size_usd_next2year with Controls:, no clustering

	Independent_Variable	Coefficient	Std_Error	T_Stat	P_Value	[0.025	0.975]	R_Squared	Adj_R_Squared	
0	SRR_median	0.638849	0.424394	1.505321	0.132573	-0.194007	1.471704	0.075443	0.063789	6
1	SRR_mean	0.607953	0.436309	1.393400	0.163824	-0.248286	1.464192	0.075128	0.063470	6
2	normalized_SRR_mean	-15.125348	72.947316	-0.207346	0.835784	-158.281465	128.030768	0.073284	0.061603	6
3	normalized_SRR_median	-13.940456	72.921235	-0.191171	0.848432	-157.045388	129.164477	0.073278	0.061596	6

SRR normalized is not significant in next 2 year transaction model

Regression with Controls Results on log_transaction_size_usd_next2year (clustered by gvkey):

	Independent_Variable	coef	Std_Error	T_Stat	P_Value	[0.025	0.975]	R_Squared	Adj_R_Squared	N
0	SRR_median	0.638849	0.405720	1.574605	0.115348	-0.156348	1.434045	0.075443	0.063789	
1	SRR_mean	0.607953	0.425424	1.429051	0.152990	-0.225863	1.441769	0.075128	0.063470	
2	normalized_SRR_mean	-15.125348	53.067998	-0.285018	0.775630	-119.136713	88.886016	0.073284	0.061603	
3	normalized_SRR_median	-13.940456	53.066138	-0.262700	0.792782	-117.948174	90.067263	0.073278	0.061596	



1. Winsorize transaction size

Trim the transaction size distribution by excluding observations below the 2nd percentile and above the 98th percentile

- normalized_SRR is positively and significantly associated with contemporaneous acquisition size. The estimate remains stable and statistically significant after controlling for firm and CEO characteristics and when including sector fixed effects

2. Check multicollinearity

Method: variance inflation factors (VIFs)

High VIF values indicate strong multicollinearity, which can increase standard errors and produce unstable coefficient estimates across specifications.

- Variance inflation factors (VIFs) for the main explanatory variable and all controls are low (all < 2.1), indicating that multicollinearity is not a material concern

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1. Simple:      coef=128.932  se=58.944  p=0.0287
2. + Controls:  coef=169.829  se=67.840  p=0.0123
3. + Sector FE: coef=174.912  se=62.522  p=0.00515
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```

	feature	VIF
0	normalized_SRR_mean	1.008127
1	TobinQ	1.472676
2	tenure_years	1.788360
3	gender	1.055481
4	CF	1.291133
5	age	1.524452
6	Aqc_Ratio	1.083126
7	shrown_excl_opts_pct	2.076961
8	revt_lag	1.098426
9	Leverage	1.045056
10	log_option_awards	1.142242
11	log_stock_awards	1.361907
12	opt_unex_exer_est_val	1.434540



1

SRR is statistically significant and positively associated with the acquisition activity next year

2

Transaction magnitude is predominantly governed by firm-level capacity and market positioning rather than individual executive traits.

3

A firm's size (existing balance sheet capacity) is a strong anchor for future deal size.

4

Highly profitable firms leverage their superior cash-flow generation to execute larger-scale acquisitions.

5

Firms with high market-to-book valuations—often interpreted as having a "mandate for growth"—consistently pursue larger targets.



- [1] Liu, Yue (Lucy) and Taffler, Richard J., Damned Out of Their Own Mouths: CEO Narcissism in M&A Decision-Making and Its Impact on Firm Performance (February 15, 2009). Available at SSRN: <https://ssrn.com/abstract=1344119> or <http://dx.doi.org/10.2139/ssrn.1344119>

